THE HARTFORD- HCL GUVI: TRAINING PROGRAM FOR .NET FULL STACK

Audience: Tier-1 College Graduate Developers & Engineers
Duration : 8 Hours per day : 11 Weeks (including Soft Skills programme)
Delivery Mode: Theory + Live Demos + Hands-on Practice

- Instructor: Industry Mentors
- Session-wise Gamified Quiz challenges (for better interactions & knowledge sharing)
- Hands-on practice with use cases

Week 1 : Foundation Bootcamp

Day 1 – SQL Server Basics & Data Definition

Module 1.1: Introduction to SQL Databases

- What is a SQL Database
- SQL Server architecture overview
- Types of SQL Servers (Express, Standard, Enterprise)

Module 1.2: Exploring SQL Server Sample Database – Live Demo

- Installing/connecting to SQL Server
- Exploring AdventureWorks / sample databases
- Overview of tables, views, relationships

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Practice Session 1: Exploring Sample Database

- Query tables: SELECT, WHERE, ORDER BY
- Examine table structures and sample data

Module 1.3: Data Definition Language – DDL – Theory + Demo

- CREATE, ALTER, DROP, TRUNCATE
- SELECT INTO, RENAME, SYNONYM

Practice Session 2: DDL Exercises

- Create tables for a simple Insurance database
- Alter table schema (add/remove columns)
- Create synonyms and SELECT INTO new tables

Module 1.4: Constraints – Theory + Demo

- PRIMARY KEY, FOREIGN KEY
- NOT NULL, UNIQUE, CHECK constraints

Practice Session 3: Constraints Exercise

- Add PK/FK to Insurance tables
- Apply constraints to ensure data integrity

Daily Deliverable: Sample Insurance database created with tables, constraints, and initial sample data.

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Day 2 – Querying & Modifying Data

Module 2.1: Querying Data – Theory + Demo

- SELECT, filtering (WHERE, LIKE, IN, BETWEEN)
- Sorting data (ORDER BY)
- Limiting rows (TOP / OFFSET-FETCH)

Practice Session 1: Querying Data

- Filter and sort insurance policies by type and premium

Module 2.2: Modifying Data – Theory + Demo

- INSERT, UPDATE, DELETE, MERGE

Practice Session 2: Data Modification

- Insert new policies
- Update premiums
- Delete canceled policies

Module 2.3: Joining Tables – Theory + Demo

- INNER, LEFT, RIGHT, FULL OUTER, CROSS, SELF JOIN

Practice Session 3: Joins Exercise

- Query policies with customer info
- List all vehicles with coverage details

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Module 2.4: Grouping Data – Theory + Demo

- GROUP BY, ORDER BY, GROUPING SETS, CUBE, ROLLUP

Daily Deliverable: Queries retrieving meaningful insurance data with joins, grouping, and updates.

Day 3 – Subqueries, Set Operators & CTEs

Module 3.1: Subqueries – Theory + Demo

- Simple subqueries, correlated subqueries
- EXISTS, ANY, ALL

Practice Session 1: Subquery Exercise

- Find customers with multiple policies
- Check policies with high premiums

Module 3.2: Set Operators – Theory + Demo

- UNION, INTERSECT, EXCEPT

Practice Session 2: Set Operators Exercise

- Compare policies across different insurance types

Module 3.3: Common Table Expressions (CTE) – Theory + Demo

- CTE & Recursive CTE
- Pivot tables

Practice Session 3: CTE Exercise

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

- Create policy summary using CTE
- Pivot policy counts by type and region

Module 3.4: Expressions & Views – Theory + Demo

- CASE, COALESCE, NULLIF
- SQL Server Views creation and usage

Daily Deliverable: Advanced queries using subqueries, CTEs, set operators, and expressions.

Day 4 – Indexing, Performance & Practical SQL Project

Module 4.1: Indexes – Theory + Demo

- Clustered vs Nonclustered
- Rename, Enable, Disable, Drop, Filtered indexes

Practice Session 1: Index Exercise

- Create indexes for Insurance tables
- Optimize query performance

Module 4.2: Useful Tips – Theory + Demo

- Find duplicates
- Delete duplicates

Module 4.3: Performance Tuning – Theory + Demo

- Query analysis: WHERE vs JOIN

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

- Using constraints
- Execution plan, logical I/Os

Practice Session 2: Performance Tuning

- Analyze queries on Insurance DB
- Optimize slow queries using indexes and execution plan

Module 4.4: Practical Project Assignment

- Model Insurance Company DB (house, auto, family coverage, vacation home, boat, motorcycle)
- Design schema with tables, constraints, and sample data

Daily Deliverable: Optimized insurance database with indexes and queries ready for use.

Day 5 – MongoDB (NoSQL) Fundamentals & Project

Module 5.1: Introduction & CRUD Operations – Theory + Demo

- DB, Collection, Document
- CRUD vs CRUD Many
- Projection

Practice Session 1: MongoDB CRUD

- Create collections for insurance
- Insert multiple documents
- Retrieve data with projections

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Module 5.2: Embedded Documents & Arrays – Theory + Demo

- Embedded vs referenced documents
- Working with arrays

Practice Session 2: Embedded Data

- Store policy coverage options in embedded documents
- Query arrays for customer policies

Module 5.3: Schema Types & Relationships – Theory + Demo

- Data modeling in MongoDB
- Embedded vs reference methods

Practice Session 3: MongoDB Project

- Model Insurance DB in MongoDB
- Include House, Auto, Vacation, Boat, Motorcycle policies
- Query and retrieve structured data

Module 5.4: Final Project Wrap-up

- Compare SQL vs MongoDB design
- Discuss pros and cons
- Submit project for evaluation

Daily Deliverable: Fully modeled Insurance DB in MongoDB with sample data, queries, and reports.

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Week 2 - Web Development Skills

Day 1 – HTML Basics & Project Setup

Module 1.1: Introduction to Web Development – Theory

- Application Development Life Cycle
- Front-End Dev vs Back-End Dev
- Project setup, file structure

Module 1.2: HTML Fundamentals – Theory + Demo

- Tag structure: Block, Inline & Empty tags
- Formatting tags, tables, web fonts
- Images, forms, iframes, video & audio

Practice Session 1

- Create a simple webpage with headings, paragraphs, images, table, form elements

Module 1.3: HTML Project Assignment – Hands-on

- Build **Bento Grid Page**: Frontend Mentor Bento Grid

Module 1.4: Introduction to CSS – Theory + Demo

- CSS introduction, selectors, applying styles
- Reset CSS, formatting styles, backgrounds

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Practice Session 2

- Style HTML page created in Practice Session 1: colors, fonts, borders, backgrounds

Daily Deliverable: Bento Grid webpage with basic styling applied

---

Day 2 – CSS Box Model, Flexbox & Grid

Module 2.1: CSS Layouts – Theory + Demo

- Box Model, Margin, Padding, Border
- Flexbox basics, alignments, justify content

Practice Session 1

- Create a responsive flexbox layout for a simple page

Module 2.2: CSS Grid – Theory + Demo

- Grid container, items, columns, rows
- Grid templates, gaps, alignment

Practice Session 2

- Build grid-based product list layout: Frontend Mentor Product List

Module 2.3: Media Queries & Responsiveness

- Breakpoints, responsive design principles

Practice Session 3

- Make Product List page responsive for mobile, tablet, desktop

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Module 2.4: Tailwind CSS Introduction – Theory + Demo

- How Tailwind works, utility-first approach
- Setting up Tailwind, installing via npm

Daily Deliverable: Responsive product list page using CSS and Flex/Grid, basic Tailwind setup

---

Day 3 – Tailwind Components & Layouts

Module 3.1: Tailwind Fundamentals

- Using wiki & prebuilt components
- Utility classes: spacing, typography, colors

Practice Session 1

- Change background, padding & margins using Tailwind

Module 3.2: Layouts & Components

- Creating Nav Bar, Gallery, Placement & sizing of elements

Practice Session 2

- Create responsive nav bar and image gallery

Module 3.3: Grids & Flexbox with Tailwind

- Flexbox alignment, grid layouts, customizing Tailwind config

Practice Session 3

- Replicate e-commerce product page with Tailwind: Frontend Mentor E-commerce

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Module 3.4: Practical Tailwind Assignment

- Practice customizing Tailwind, adding responsive layouts

Daily Deliverable: Fully responsive Tailwind-based e-commerce product page

---

Day 4 – JavaScript Fundamentals & DOM

Module 4.1: JavaScript Basics

- Variables, data types, operators
- ES6 features: let/const, template literals, arrow functions

Practice Session 1

- Manipulate page content with variables and template literals

Module 4.2: Problem Solving using JS

- map, filter, reduce, loops
- Writing clean, standard code (prettier, eslint)

Practice Session 2

- **Vibe Coding :** Implement logic for Bento Grid page using JS array methods

Module 4.3: DOM Manipulation – Theory + Demo

- What is the DOM?
- Selecting elements, changing content, attributes, styles
- Traversing the DOM

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Practice Session 3

- Add interactive behavior to Bento Grid page: toggle classes, update content

Daily Deliverable: Dynamic Bento Grid page with JS interactivity

---

Day 5 – Advanced JS, Events & Final Project

Module 5.1: DOM Events & Event Listeners

- Event bubbling, capturing
- stopPropagation method
- Handling user interactions

Practice Session 1

- Add click, hover events on buttons, links, or images

Module 5.2: Advanced Project Assignment

- Tic-Tac-Toe game using HTML/CSS/Tailwind/JS: Frontend Mentor Tic-Tac-Toe

Practice Session 2

- Add interactive features: player turns, win detection, reset button

Module 5.3: Project Wrap-Up & Code Review

- Review JS, Tailwind, and DOM code
- Best practices for readability and maintainability

Daily Deliverable: Fully functional Tic-Tac-Toe game with responsive layout and JS interactivity

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Week 3, Week 4: Advanced Javascript and Angular

Day 1 – Advanced JavaScript Foundations

Module 1.1 – Callbacks & Callback Hell
- Async vs sync execution
- Callback structure and control flow
- Problems with nested callbacks and callback hell

Module 1.2 – Promises & Chaining
- Promise states and executor
- .then(), .catch(), .finally()
- Error propagation and chaining

Practice 1 – Convert nested callbacks into Promise chains

Module 1.3 – Async/Await & Error Handling
- Async/await syntax and benefits
- try/catch and rejection handling
- Parallel vs sequential awaits

Practice 2 – Refactor Promise chains using async/await

Module 1.4 – Event Loop & Task Queues
- Call stack vs heap
- Microtasks vs macrotasks
- How Promises and setTimeout are scheduled

Practice 3 – Visualize Event Loop with console.log/setTimeout/Promise demo

---

Day 2 – REST APIs & Postman

Module 2.1 – What is a REST API
- HTTP methods (GET, POST, PUT, DELETE)
- Request/response structure and status codes

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Module 2.2 – Postman Essentials
- Setup and send first API call
- Collections, variables, and authorization

Practice 1 – Play with CRUD APIs using mockapi.io in Postman

Module 2.3 – CORS and Client/Server Rendering
- CORS policy and preflight requests
- Why CSR vs SSR and how they differ

Practice 2 – Submit Guvi.in Contact Form data to mockapi.io

Module 2.4 – Mini Project Setup
- Initialize Coffee Roasters site (HTML/CSS/JS/Tailwind scaffold)

Day 3 – TypeScript Fundamentals

Module 3.1 – TypeScript Basics
- Why TypeScript over JavaScript
- Types, compilation, and tsconfig

Module 3.2 – Classes, Interfaces & Enums
- Class syntax, inheritance, access modifiers
- Interfaces and enums

Practice 1 – Model Coffee Roasters data with interfaces and enums

Module 3.3 – Generics and Utility Types
- Generic functions and classes
- Built-in utility types (Partial, Pick, Readonly)

Practice 2 – Create generic API helpers for CRUD operations

Day 4 – Angular Introduction

Module 4.1 – What is Angular & Why a Framework
- AngularJS vs Angular 20

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

- SPA architecture and benefits
- Ivy compiler overview

Module 4.2– Environment Setup
- Install Node, NPM, Angular CLI
- Create and run first Angular app

Module 4.3 – Project Structure & Workflow
- package.json vs package-lock.json
- Versioning and dependency resolution

Practice 1 – Build “Hello Angular” SPA

Day 5 – Components & Architecture

Module 5.1 – MVC in Angular
- Model, View, Controller pattern
- Component as the core unit

Module 5.2 – Creating Components
- HTML template (view), TS file (model), CSS (styling)
- Component decorator and metadata

Practice 1 – Create Header and Footer components

Module 5.3– Filters & Animations
- Built-in pipes and custom pipes
- ngAnimate, CSS transitions and transforms

Practice 2 – Animate Coffee Roasters cards on load

Day 6 – Data Binding & Services

Module 6.1 – Lifecycle Hooks
- ngOnInit, ngOnChanges, ngDestroy and their uses

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Module 6.2 – Data Binding Techniques
- String interpolation, property binding, event binding
- Two-way binding with ngModel

Practice 1 – Bind form data in subscription form

Module 6.3 – Services & HttpClient
- Creating and injecting services
- Fetching data from API with HttpClient

Practice 2 – Create ProductService to fetch Coffee Roasters data

Mini Project: Integrate API data into Coffee Roasters page

Day 7 – Standalone Components & Forms

Module 7.1 – Standalone Components vs NgModule
- Concept and advantages

Practice 1 – Refactor traditional component into standalone

Module 7.2 – Reactive Forms
- FormGroup, FormControl, FormArray
- Validation patterns

Practice 2 – Reactive Login Form with custom validators

Module 7.3 – Passing Data between Components
- @Input, @Output decorators
- Service-based communication

Daily Deliverable: Functional login/registration flow using standalone components

---

Day 8 – Modules, Directives & Signals

Module 8.1 – AppModule & Routing Basics
- AppComponent, AppRoutingModule, dependencies

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Module 8.2 – Directives & Pipes
- Structural (*ngFor, *ngIf) vs Attribute directives
- Create a custom directive and pipe

Module 8.3 – Signals in Angular 20
- Signals vs Observables
- Derived and Computed Signals

Practice 1 – Reactive Counter using Signals
Mini Project: To-Do List app using Signals for state management

---

Day 9 – Testing & Debugging

Module 9.1 – Unit Testing with Jasmine & Karma
- Angular TestBed, mocking services
- Writing test suites and spec files

Practice 1 – Test Reactive Form validation logic

Module 9.2 – Component Testing & Debugging
- User Badge Component testing
- Chrome DevTools for Angular debugging

Practice 2 – Simulate slow API response and optimize performance

---

Day 10 – Routing, Lazy Loading & Angular 20 Features

Module 10.1 – Routing Advanced
- Lazy Loading, Route Guards, Resolvers

Practice 1 – Implement lazy-loaded routes for Courses/Login/Dashboard

Module 10.2 – Styling Frameworks
- Integrate Tailwind or Bootstrap with Angular

Module 10.3– What's New in Angular 20
- Signal-based APIs, zoneless change detection, SSR hydration, new builder

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Final Project – Replicate **www.guvi.in** Home + Courses Pages
• Tailwind + Angular 20 features
• Routing between pages

Week 5 : .NET with C#

Day 1: Introduction to .NET Framework/Core and C# Basics

Module 1.1: Introduction to .NET Framework/Core

- Overview of .NET Framework and .NET Core
- CLR, CTS, CLS, GC, JIT Compiler, Base Class Libraries
- Managed and Unmanaged code

Module 1.2: Setting up .NET Core Environment

- Acquiring and installing .NET Core SDK
- What's new in .NET Core 6
- Creating a console application
- Debugging and publishing a console app

Module 1.3: Class Library Basics

- Create a class library
- Unit testing a class library

Practice 1:

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

- Create a simple console app (“Hello World”)
- Debug the app and handle exceptions
- Create a class library and write a simple unit test

Day 2: C# Program Structure and Data Types

Module 2.1: Basics of C# Programming

- What is C#? Program structure explained
- Hello World explained
- Data types, variables, and operators

Module 2.2: Strings and Memory Management

- Basics of strings
- Searching and manipulating strings
- Garbage collection, Finalize, Dispose

Module 2.3: Windows Forms Overview

- Introduction to Windows Forms applications

Practice 2:

- Implement string operations and search functionality
- Create a basic Windows Forms application with simple input and output

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

- Demonstrate proper disposal of resources

Day 3: Control Flow and Methods

Module 3.1: Control Flow Statements

- Branches: if, else if, else
- Loops: for, while, do-while
- Combining branches and loops for complex logic

Module 3.2: Methods and Functions

- Defining and calling methods
- Passing parameters by value and reference
- Returning values and method overloading

Module 3.3: Numbers and Math

- Integer and decimal precision
- Number operations, rounding, and conversions

Practice 3:

- Implement a console program using branches and loops
- Write methods for common calculations and validations
- Solve algorithmic problems using control flow and recursion

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Day 4: Arrays, Collections, Generics, and Indexers

Module 4.1: Arrays and Lists

- Creating and accessing arrays
- Lists and collections overview
- Generic types and benefits

Module 4.2: Properties and Indexers

- Using properties to encapsulate data
- Creating indexers for classes

Module 4.3: Sorting, Searching, and Collection Operations

- Sort, search, and index lists
- Working with lists of different types

Practice 4:

- Implement an employee collection using List
- Add sorting and searching operations
- Use properties and indexers for accessing employee data

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Day 5: Objects, Classes, Inheritance, Polymorphism, and Clean Code Principles

Module 5.1: Object-Oriented Programming in C#

- Classes and objects
- Associations between classes
- Inheritance and polymorphism
- Interfaces, methods, and members

Module 5.2: Principles of Clean Code and Naming

- Naming conventions for variables, classes, and enums
- Communicating intent through names
- Avoiding disinformation, encoding, and subtle variations
- Boolean variables and predicates, naming conventions for properties and methods

Practice 5:

- Implement an object-oriented employee management system
- Use inheritance, interfaces, and polymorphism
- Apply clean code naming principles
- Solve algorithmic problems integrating classes, collections, and recursion

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Week 6 to Week 8 : Advanced C#, .NET, Agile, DevOps

Day 1: Exceptions and Delegates

Module 1.1: Exceptions in C#

- Exception handling overview
- Try-catch-finally blocks
- Throwing and creating custom exceptions

Module 1.2: Delegates

- Single-cast and multi-cast delegates
- Syntax and use cases

Module 1.3: Lambda Expressions

- Lambda syntax and use with delegates

Practice 1:

- Implement a console application to demonstrate exception handling
- Create delegates and invoke single/multi-cast delegates using lambda expressions

---

Day 2: Events, Extension Methods, and LINQ

Module 2.1: Events

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

- Event declaration and handling
- Event-driven programming

Module 2.2: Extension Methods

- Creating and using extension methods

Module 2.3: LINQ Basics

- LINQ queries with collections

Practice 2:

- Build an Employee database using collections
- Implement LINQ queries to search and filter data
- Apply exception handling for invalid inputs

Day 3: Nullable Types, Dynamic, and Asynchronous Programming

Module 3.1: Nullable Types and Dynamic / var

- Nullable value types
- Using dynamic and var for flexibility

Module 3.2: Asynchronous Programming

- async and await keywords

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

- Writing asynchronous methods for IO-bound tasks

Practice 3:

- Implement asynchronous data retrieval for Employee records
- Apply nullable types and dynamic variables where appropriate

---

Day 4: Introduction to REST APIs and .NET Core API Structures

Module 4.1: REST API Basics

- API design principles
- HTTP methods, status codes, request/response structure

Module 4.2: .NET Core API Project Structure

- Controllers, Models, Services, Repositories
- Dependency Injection and object lifecycle

Module 4.3: Swagger UI

- Documentation of APIs using Swagger

Practice 4:

- Create a sample REST API project with basic CRUD operations
- Use Swagger to document endpoints

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Day 5: Middleware and API Filters

Module 5.1: Middleware in .NET Core

- Request and response pipeline
- Custom middleware

Module 5.2: API Filters

- Action filters, exception filters
- Logging and validation

Module 5.3: API Testing

- Unit testing controllers
- Integration testing APIs

Practice 5:

- Build CRUD APIs for Employee management
- Implement middleware for logging
- Apply filters for input validation

Day 6: Practical Task Assignment 1 – Employee Database Project

Module 6.1: Project Overview

- Collection-based Employee database

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

- Search operations and validation

Practice 6:

- Implement console app with OOP principles
- Handle exceptions properly
- User-friendly error messages

Day 7: Entity Framework Core and ORM

Module 7.1: EF Core Basics

- Setting up EF Core
- DbContext, DbSet, and models

Module 7.2: CRUD with ORM

- Adding, updating, deleting, querying data using EF Core

Module 7.3: Authentication and Authorization (JWT)

- Implement JWT-based authentication for APIs

Practice 7:

- Implement CRUD operations using EF Core
- Secure API endpoints using JWT

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Day 8: Multithreading and Parallel Programming Basics

Module 8.1: Multithreading & Concurrency

- Thread creation, thread safety, synchronization

Module 8.2: Parallel Programming

- Parallel loops and tasks
- Task Parallel Library (TPL)

Module 8.3: Code Complexity

- Measuring cyclomatic and cognitive complexity

Practice 8:

- Implement parallel processing for batch employee data operations
- Analyze code complexity of methods

Day 9: SOLID Principles and Clean Architecture

Module 9.1: SOLID Principles

- Single Responsibility, Open/Closed, Liskov Substitution, Interface Segregation, Dependency Inversion

Module 9.2: Applying SOLID to APIs

- Structuring services, repositories, and controllers

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Practice 9:

- Refactor Employee API to follow SOLID principles
- Apply interface segregation and dependency injection

Day 10: Practical Task Assignment 2 – Fullstack Employee Project

Module 10.1: Project Setup

- Backend with .NET Core API and SQL Server
- Frontend with Angular and Tailwind UI

Module 10.2: Authentication & Authorization

- Implement JWT-based login
- Role-based access control

Practice 10:

- Full CRUD operations for Employee management
- Secure API endpoints and connect frontend to backend

Day 11: Advanced REST API Practices

Module 11.1: Middleware and Filters

- Logging, exception handling, validation

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Module 11.2: Testing REST APIs

- Unit tests for controllers
- Integration tests

Module 11.3: API Documentation

- Swagger enhancements and versioning

Practice 11:

- Enhance Employee API with middleware and filters
- Test all endpoints thoroughly

Day 12: Best Practices and Fullstack Demonstration

Module 12.1: API Development Best Practices

- Error handling, logging, response standardization
- Versioning, caching, pagination

Module 12.2: Fullstack Demonstration

- Angular frontend integration with .NET Core API
- Using Tailwind UI for styling

Practice 12:

- Complete a mini fullstack project with Angular & .NET

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

- Implement proper error handling and validations

Day 13: Practical Task Assignment 3 – Fullstack Project with ORM & Authentication

Module 13.1: Project Requirements

- Backend with ORM (EF Core) and SQL Server
- Frontend with Angular + Tailwind UI
- JWT-based authentication and authorization

Practice 13:

- Implement full CRUD functionality
- Connect frontend to backend with proper API calls
- Ensure security and validations

Day 14: Agile Methodology and Project Management

Module 14.1: Agile Principles and Practices

- Scrum and Kanban frameworks
- Sprint planning, daily stand-ups, sprint reviews, retrospectives

Practice 14:

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

- Plan tasks for mini-project using Agile principles
- Conduct a mock sprint review and retrospective

Day 15: Cloud Computing & DevOps Overview

Module 15.1: Cloud Computing Concepts

- Benefits of cloud computing
- Service models: IaaS, PaaS, SaaS
- Deployment models: Public, Private, Hybrid

Module 15.2: Azure Services Overview

- Azure regions, availability zones
- Compute: VMs, App Services, Containers
- Storage: Blob, Disk, File, Queue
- Databases: Azure SQL, Cosmos DB

Module 15.3: DevOps & CI/CD

- Introduction to Azure DevOps
- Creating pipelines, builds, releases
- Deploying fullstack application

Practice 15:

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

- Deploy fullstack Employee project to Azure
- Configure CI/CD pipeline for automated builds and releases

Week 9: AI for Developers

Day 1: AI-Powered Coding with GitHub Copilot

Module 1.1: AI-Assisted Development (60 min) - Theory

- How GitHub Copilot works: context windows and code generation
- Setup in VS Code and JetBrains
- Security and responsible AI coding practices
- Productivity gains in enterprise development

Module 1.2: Copilot in Action (60 min) - Live Demo

- Code generation: inline suggestions, multi-line completions
- Using Copilot Chat for debugging and explanations
- Natural language to code conversion
- Demo: Building a REST API from scratch

Practice Session 1: Build Claims API (120 min)

- **Use Case:** Insurance Claims Submission API
- **Build:** 3 endpoints - create claim, get claim, update claim
- **Add:** Basic validation

Module 1.3: Advanced Copilot Techniques (45 min) - Theory

- Test-driven development with AI
- Refactoring legacy code with Copilot
- Documentation generation

Practice Session 2: Add Tests & Documentation (75 min)

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

- **Use Case:** Enhance Claims API
- **Build:** Unit tests with Copilot
- **Generate:** API documentation

Daily Deliverable: Functional Claims API with tests

Day 2: Document Intelligence with Gemini

Module 2.1: Google Gemini Fundamentals (45 min) - Theory

- Gemini model family: 1.5 Pro, 1.5 Flash, 2.0 Flash
- Multimodal AI: text, images, PDFs processing
- API vs Vertex AI deployment options

Module 2.2: Document Processing (75 min) - Live Demo

- PDF and image analysis for business documents
- Structured data extraction from unstructured sources
- OCR and handwriting recognition
- Demo: Extracting claim information from documents

Practice Session 1: Basic Document Extractor (90 min)

- **Use Case:** Process Insurance Claim Form
- **Build:** Upload PDF, extract key fields (name, policy #, date)
- **Output:** JSON format

Module 2.3: Processing Pipelines (30 min) - Theory

- Batch processing and async operations
- Error handling and confidence scoring

Practice Session 2: Enhanced Document Processor (150 min)

- **Use Case:** Multi-Document Processing
- **Build:** Process 2-3 document types
- **Add:** Validation checks, simple web UI

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Daily Deliverable: Document processing pipeline with basic UI

Day 3: Agentic AI with LangChain and Langraph

Module 3.1: Agentic AI Fundamentals (60 min) - Theory

- AI agents vs chatbots
- Agent architectures: ReAct, Plan-and-Execute
- Components: Memory, Tools, Planning, Execution

Module 3.2: LangChain Essentials (60 min) - Live Demo

- Architecture: Models, Prompts, Chains, Agents, Memory
- Prompt engineering for deterministic outputs
- Memory types
- Demo: Building a Q&A agent

Practice Session 1: Simple Q&A Agent (90 min)

- **Use Case:** Policy Information Bot
- **Build:** Answer questions about insurance policies
- **Add:** Conversation memory

Module 3.3: Tools & Function Calling (45 min) - Demo

- Creating custom tools for agents
- Function calling with Gemini
- Integrating external APIs

Practice Session 2: Agent with Tools (135 min)

- **Use Case:** Policy Assistant with Database
- **Build:** Query mock policy data
- **Add:** 2 custom tools (database query, calculator)

Daily Deliverable: Agent with tool integration

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Day 4: Advanced Agentic AI with Vertex AI

Module 4.1: Vertex AI Platform (60 min) - Theory

- Vertex AI ecosystem: AI Studio, Agent Builder, Model Garden
- Model deployment and management
- RAG patterns

Module 4.2: Vertex AI Agent Builder (60 min) - Live Demo

- Agent Builder UI and workflow configuration
- Creating data stores
- Demo: Building a simple agent

Practice Session 1: Deploy Agent on Vertex AI (90 min)

- **Use Case:** Claims Status Lookup
- **Build:** Single agent on Vertex AI
- **Deploy:** Basic API endpoint

Module 4.3: Multi-Agent Systems with LangGraph (30 min) - Theory

- When to use multiple agents
- LangGraph for agent workflows and state management
- Agent communication patterns and handoffs
- Building graphs with nodes and edges

Practice Session 2: Two-Agent System with LangGraph (150 min)

- **Use Case:** Simple Claims Pipeline
- **Build:** Agent 1 validates input, Agent 2 processes using LangGraph
- **Add:** State management and agent handoff

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Day 5: Production Deployment & Capstone

Module 5.1: Production Best Practices (60 min) - Theory

- Security: PII handling, data privacy
- Model monitoring basics
- Cost optimization
- Safety guardrails

Module 5.2: Cloud Deployment (60 min) - Live Demo

- Containerization with Cloud Run
- Basic authentication
- Demo: Deploying to production

Practice Session 1: Deploy Previous Work (60 min)

- **Use Case:** Production Deployment
- **Build:** Containerize Day 4 agent
- **Deploy:** To Cloud Run with basic auth

Module 5.3: Legacy Integration (30 min) - Theory

- AI-assisted code documentation
- Integration patterns

Capstone Project (240 min)

Choose ONE:

Option A: Claims Submission System

- Accept claim form
- Extract data with Gemini
- Store in database
- Return confirmation

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Option B: Policy Q&A Bot

- Answer policy questions
- Use database for policy info
- Provide relevant responses
- Track conversation history

Option C: Document Validator

- Upload insurance documents
- Check for required fields
- Flag missing information
- Generate validation report

Final Session: Presentations (60 min)

- Team demonstrations
- Feedback
- Certification

Week 10 : C2C (Softskills) Programme

<< Kindly refer to Annexure document for C2C Curriculum >>

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI

Week 11: Capstone Project

- Build a .NET Core REST API to manage employee data with add, view, update, and delete operations.
- Secure the API using login and role-based access.
- Create a web interface with Angular and Tailwind to input employee details and view lists.
- Connect the frontend seamlessly with the backend API.
- Use SQL Server to store employee data and manage it through Entity Framework Core.
- Implement error handling with friendly messages.
- Add search, filter, and sorting features for employee records.
- Follow good coding practices including SOLID principles and clean code.
- Deploy the fullstack project to Azure cloud.
- Set up a CI/CD pipeline for automated deployment.

© 2025 GUVI GEEK NETWORK PVT LTD. All rights reserved. No part of this document may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior permission from GUVI